

Technical Specifications, Budget & Certifications

Comprehensive Engineering, Financial & Institutional Verification Dossier

Applicant: Mubarak Raji Babatunde (Project Leader) | Lagos State University (LASU) | CAC BN-9846744

Section 1: Hardware Engineering & BOM Specifications

Detailed Bill of Materials, component costs (\$15 total BOM), electrical pinouts (ESP32, ADXL345, DS18B20, INMP441), and magnetic enclosure CAD specs.

Section 2: 12-Month Milestone Budget Allocation (NGN 50M)

Phase-by-phase financial expenditure breakdown across R&D, pilot SMT PCB manufacturing, fleet onboarding, cloud infrastructure, and operations.

Section 3: Software & AI Telemetry Architecture

Retrieval-Augmented Generation (RAG) manual indexing pipeline, WhatsApp bot webhook flow, PostgreSQL/pgvector database schema, and 75% benchmark accuracy.

Section 4: Institutional Endorsement & Competition Nomination

The Nigerian Society of Engineers (NSE, Lagos Branch) official invitation letter to LASU regarding the 15th China Innovation & Entrepreneurship Competition (AI + Construction Machinery).

Section 5: Professional Credentials & Certifications

Verified certifications for team members including Data Analysis, SolidWorks Mechanical Design (CSWP), and Interswitch Hackathon recognition.

1. Hardware Specifications & Bill of Materials (BOM)

Component	Function / Specification	Unit Cost (NGN)	Unit Cost (USD)
ESP32-WROOM-32	240MHz Dual Core MCU, Wi-Fi & BLE, Edge FFT	NGN 5,500	\$3.50
ADXL345 Accelerometer	3-Axis Digital Vibration (+-16g, SPI/I2C)	NGN 2,800	\$1.75
DS18B20 Temp Probe	Waterproof Stainless Probe (-55C to +125C)	NGN 1,800	\$1.15
INMP441 Microphone	Omnidirectional I2S Digital MEMS Acoustic	NGN 2,200	\$1.40
18650 Li-ion Battery	3.7V 2600mAh Rechargeable Cell	NGN 2,500	\$1.60
TP4056 + Protection	Li-ion Charging Module with Micro-USB/5V	NGN 800	\$0.50
Neodymium Magnets	5x N52 Heavy Duty Magnets (50kg pull force)	NGN 2,500	\$1.60
3D-Printed Enclosure	High-Temp Heat-Resistant PETG/PLA Shell	NGN 3,500	\$2.20
Wiring & Passives	Silicone Wires, Resistors, Switch, Screws	NGN 1,500	\$1.00
TOTAL UNIT COST	Complete Assembled MechMind Node v1	NGN 23,100	\$14.70

2. Operational Specifications & Edge Telemetry

- Vibration Sampling:** 1600 Hz sampling rate, 512-point FFT computed on ESP32, RMS acceleration.
- Thermal Monitoring:** Surface temperature sampled every 5 seconds, alarm threshold configurable (+85C).
- Acoustic Telemetry:** I2S digital audio streaming with frequency band energy calculation.
- Mounting Mechanism:** 5x Neodymium disc magnets clamped directly onto excavator engine block or hydraulic pump.
- Battery Longevity:** Up to 36 hours continuous monitoring; extends to 14 days in sleep-telemetry cycle mode.

12-Month Milestone-Based Budget Allocation (NGN 50,000,000)

Phase 1 (Months 1–3): Pilot Production & Lab Validation • Component procurement & fabrication of 50 SMT PCB sensor nodes: NGN 3,500,000 • Heavy equipment test-rig calibration & lab bench test instruments: NGN 3,000,000 • Cloud GPU & pgvector database infrastructure setup: NGN 2,500,000 • Manual indexing pipeline & RAG data expansion (20 models): NGN 2,000,000 • Engineering stipends & project administration: NGN 1,500,000	NGN 12,500,000	25%
Phase 2 (Months 4–6): Fleet Deployment & WhatsApp Scaling • Field deployment of 50 sensor nodes across 10 commercial construction fleets: NGN 4,500,000 • GSM/4G cellular fallback module integration for off-grid sites: NGN 3,500,000 • WhatsApp Business API enterprise volume licensing (Twilio/Meta): NGN 2,500,000 • Mechanic onboarding & field training workshops across Lagos & Ogun: NGN 2,500,000 • Spare parts supplier API integration: NGN 2,000,000	NGN 15,000,000	30%
Phase 3 (Months 7–9): Regional Expansion & Dealer Partnerships • Scaling to 150 hardware nodes and 35 equipment brands (SANY, Komatsu): NGN 5,000,000 • Regional field expansion into Abuja & Port Harcourt infrastructure sites: NGN 3,500,000 • Equipment dealership integration & inventory sync: NGN 2,000,000 • Legal, IP filing & product safety compliance certifications: NGN 2,000,000	NGN 12,500,000	25%
Phase 4 (Months 10–12): Commercialization & Self-Sustainability • Mass assembly run of 250 units (commercial fleet supply): NGN 4,000,000 • Enterprise predictive fleet dashboard with Remaining Useful Life (RUL): NGN 2,500,000 • Customer success, field technical support & warranty reserves: NGN 2,000,000 • Final performance auditing, investor reporting & demo day: NGN 1,500,000	NGN 10,000,000	20%
TOTAL GRANT REQUEST	NGN 50,000,000	100%

AI Architecture, RAG Pipeline & Benchmark Validation

1. Multi-Modal WhatsApp Gateway

Built with Node.js and Python FastAPI using Baileys / Twilio WhatsApp API. Accepts natural language text, voice audio (transcribed via OpenAI Whisper API), and JPEG photographs of equipment nameplates or broken mechanical components.

2. Retrieval-Augmented Generation (RAG) Engine

Indexed over 6 comprehensive OEM service manuals (Caterpillar, XCMG, Komatsu). Text is chunked with semantic boundaries, embedded using high-dimensional dense vector embeddings, and stored in pgvector / PostgreSQL for low-latency similarity retrieval.

3. Edge Telemetry Ingestion Pipeline

ESP32 hardware nodes stream telemetry packets (vibration RMS, FFT peak frequency, surface temp) via lightweight MQTT and HTTP POST endpoints to the cloud backend. When readings breach dynamic baselines, automated alert triggers push directly to the mechanic's WhatsApp.

4. Public Live Monitoring Dashboard

Next.js / Vanilla web dashboard publicly deployed at <https://mechmind-ng.vercel.app> with interactive 3D model viewer, live equipment health status charts, fault log audit trail, and automated WhatsApp alert triggers.

Benchmark Validation: 75% Diagnostic Accuracy on 56 Blind Tests

The diagnostic engine was audited across 56 realistic field fault cases collected from Nigerian machinery operators:

- Hydraulic Faults (18 cases): 83.3% accuracy — Correctly identified relief valve failure, pump cavitation, cylinder seal leak.
- Electrical & Sensor Faults (16 cases): 75.0% accuracy — Successfully decoded CAN-bus J1939 fault codes and open-circuit wiring.
- Engine & Powertrain Faults (14 cases): 71.4% accuracy — Detected injector clogging, turbocharger boost loss, and overheating causes.
- Parts Matchmaking (8 cases): 87.5% accuracy — Delivered exact OEM replacement part numbers and compatible alternatives.

Institutional Endorsement: Nigerian Society of Engineers (NSE)

Official letter from NSE Lagos Branch to Lagos State University endorsing the 15th China Innovation & Entrepreneurship Competition (XCMG AI + Construction Machinery)



The Nigerian Society of Engineers
(LAGOS BRANCH)
Motto: Striving for Engineering Excellence
Oyebolu Engineering Resource Centre By Commercial Road, Works & Physical Planning Department
University of Lagos Akoka, Yaba.
E-mail: nseifsecretariat@gmail.com secretariat@nse-lagos.org.ng Website: www.nse-lagos.org.ng

CHAIRMAN
ENGR. BOLAJI ADEBAJO, FNSE
0803 208 4127

GENERAL SECRETARY
ENGR. MERCY ADEDAYO, MNSE
0802 391 0384

10th September ,2026

The Vice-Chancellor
Lagos State University (LASU),
Ojoo,
Lagos.

Dear Ma,

INVITATION TO PARTICIPATE IN THE 15TH CHINA INNOVATION & ENTREPRENEURSHIP COMPETITION – “AI + CONSTRUCTION MACHINERY” DIVISION.

The Nigerian Society of Engineers (NSE), Lagos Branch, is pleased to bring to the attention of the Lagos State University an exciting opportunity for students, researchers and multidisciplinary teams of the University to participate in the **15th China Innovation & Entrepreneurship Competition – “AI + Construction Machinery” Division**, being organized by XUGONG Nigeria Company Limited (XCMG).

The professional competition, themed *“Green Intelligence for a Better World,”* focuses on the innovative application of Artificial Intelligence (AI) to construction machinery, intelligent equipment and related engineering operations.

The competition provides an excellent platform for multidisciplinary university teams to develop practical and innovative AI solutions to real-world challenges in areas including:

- Equipment maintenance and predictive maintenance;
- Mining operations;
- Fleet management and optimization;
- Construction site safety;
- Spare parts management;
- Intelligent construction machinery; and
- Other related engineering and industrial applications.

In addition, XCMG West Africa is willing to support interested teams by providing relevant industry insights, application scenarios and opportunities for engagement with construction-machinery professionals, subject to applicable compliance requirements.

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Verified Professional Certifications & Hackathon Awards

1. Oni Mujeeb Ramadan — Interswitch x Enyata Beyond the Rails Hackathon Award



2. Mubarak Raji Babatunde — Professional Data Analysis Certification (ID: 1336)



Olawale Oladele — Mechanical Engineering Credentials

Dassault Systemes SOLIDWORKS & Ansys Simulation Professional Certifications

Certified SOLIDWORKS Professional (CSWP)

Dassault Systemes Credential

Demonstrates comprehensive mastery of advanced solid modeling, complex part design, assembly constraints, design modification, and manufacturing readiness.

CSWPA — Surfacing & Complex Geometry

Dassault Systemes Specialization

Specialized certification in parametric organic surfacing, surface trim/extend/knit operations, and ergonomic aerodynamic enclosures (used for MechMind Node shell).

CSWPA — Sheet Metal Design

Dassault Systemes Specialization

Expertise in sheet metal fabrication, bend allowances, stamped enclosures, and tooling design for scalable industrial production runs.

CSWPA — Weldments & Structural Framing

Dassault Systemes Specialization

Design and validation of welded frames, structural tubular profiles, and heavy equipment mounting brackets subjected to high-vibration environments.

Ansys Mechanical Associate (Structures)

Ansys Structural FEA Credential

Finite Element Analysis (FEA) proficiency in transient structural stress, harmonic vibration response, modal frequencies, and mechanical fatigue life assessment.

Ansys Fluent Associate (Fluid Dynamics)

Ansys Computational Fluid Dynamics

CFD analysis for heat dissipation, conjugate heat transfer, and thermal airflow modeling inside rugged electronic enclosures.